

## A MAXIMUM LIKELIHOOD ESTIMATION PROCEDURE WHEN MODELLING IN TERMS OF CONSTRAINTS

**G.B. Matthews:** Department of Mathematical Statistics, University of  
Natal, Durban, 4001.

and

**N.A.S. Crowther:** Department of Statistics, University of Pretoria,  
Pretoria, 0002.

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**Summary:** This paper presents a maximum likelihood (ML) estimation procedure for the parameters in a model, which is specified in terms of its implied constraints. The procedure is derived by finding the ML estimate of the mean of the exponential family subject to the constraints  $g(\mu) = 0$ , where  $g$  is a vector-valued function of  $\mu$ , satisfying the usual regularity conditions. The result is then used to find the ML estimates for the parameters in a model. As examples, the estimation procedure is utilized to estimate the parameters in loglinear and logit models, as well as a Poisson regression model.

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### 1. INTRODUCTION

Generalized linear models were introduced by Nelder and Wedderburn (1972) and subsequently much research has appeared employing these models. In this framework the observations  $y_i$  are independent, come from an exponential family and their expectation is written as a linear model using an appropriate link function. One of the computational methods for estimating the parameters in the model is the procedure of iterative reweighted least squares. This estimation procedure is implemented in the computer package GLIM, which is widely used for fitting generalized linear models.

Agresti (1990) and McCullagh and Nelder (1989) show that the iterative method of Fisher scoring reduces to the Newton-Raphson method when the canonical link function is used. Some computer packages use the Fisher scoring procedure to find the ML estimates of the parameters in the model.

Maximum likelihood estimation subject to constraints is considered by Aitchison and Silvey (1958), in the context of a random variable  $X$  whose distribution function  $F$  depends on  $s$  parameters  $\theta_1, \dots, \theta_s$ , which are not mathematically independent but satisfy  $r$  functional relationships,  $h_i(\theta_1, \dots, \theta_s) = 0$ ,  $i = 1, \dots, r$ ;  $r < s$ . The Lagrangian multiplier method is used to find the ML estimates of the parameters, which are found numerically by a process of iteration. The estimates are shown to have an asymptotic normal distribution. Silvey (1959) further discusses the Lagrangian method and the mathematical conditions for the existence of the ML estimates. Wedderburn (1974) considers generalized linear models where the systematic component of the model is defined by a set of linear constraints. A least squares solution is presented and an application to the problem of marginal homogeneity in a contingency table is discussed.

Haber and Brown (1986) consider a maximum likelihood method for estimating the expected frequencies in a contingency table subject to certain linear constraints. The ML estimates are found in the case of a multinomial sampling procedure.

This paper provides a method of finding the ML estimate for the mean vector of the exponential family, when there are constraints of the form  $g(\mu) = 0$ , where the function  $g(\mu)$  has finite first-order partial derivatives with respect to  $\mu$ . From the estimate of the mean vector the estimates of the parameters in the model are computed. This method differs from the Newton-Raphson method where the parameters are estimated first.

The major advantages of this method are that models can be easily formulated in terms of the implied constraints, which may be linear or non-linear in  $\mu$  as will be illustrated in the applications, and that the estimation procedure can be easily implemented using a matrix algebra package.

## 2. A MAXIMUM LIKELIHOOD ESTIMATION PROCEDURE

Let  $\mathbf{X}$  be a  $p \times 1$  random vector and  $\boldsymbol{\theta}$  be a  $p \times 1$  vector of parameters. The probability function for the exponential family is given by

$$(2.1) \quad p(\mathbf{x}, \boldsymbol{\theta}) = a(\boldsymbol{\theta})b(\mathbf{x})\exp\{\mathbf{x}'\boldsymbol{\theta}\}, \quad \mathbf{x} \in \mathbb{R}^p, \quad \boldsymbol{\theta} \in \mathcal{H}.$$

$$= b(\mathbf{x})\exp\{\mathbf{x}'\boldsymbol{\theta} - \kappa(\boldsymbol{\theta})\},$$

where  $\kappa(\boldsymbol{\theta}) = -\ln a(\boldsymbol{\theta})$  and  $\mathcal{H}$  is the natural parameter space for the canonical parameter  $\boldsymbol{\theta}$ . The mean vector and the covariance matrix of  $\mathbf{X}$  are given by

$$E(\mathbf{X}) = \boldsymbol{\mu} = \frac{\partial}{\partial \boldsymbol{\theta}} \kappa(\boldsymbol{\theta}) \quad \text{and} \quad \text{Cov}(\mathbf{X}) = \frac{\partial^2 \kappa(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = \left\{ \frac{\partial^2 \kappa(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right\} = \mathbf{V}.$$

The ML estimate of  $\boldsymbol{\mu}$  without any constraints can be found as follows. The log-likelihood function is

$$\ln p(\mathbf{x}, \boldsymbol{\theta}) = \ln b(\mathbf{x}) + \mathbf{x}'\boldsymbol{\theta} - \kappa(\boldsymbol{\theta}).$$

Hence

$$\frac{\partial}{\partial \boldsymbol{\theta}} \ln p(\mathbf{x}, \boldsymbol{\theta}) = \mathbf{x} - \frac{\partial}{\partial \boldsymbol{\theta}} \kappa(\boldsymbol{\theta})$$

which when set equal to zero, gives solution  $\hat{\boldsymbol{\mu}} = \mathbf{x}$ , where  $\mathbf{x}$  is a vector observation of  $\mathbf{X}$ . This is a maximum since

$$\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \ln p(\mathbf{x}, \boldsymbol{\theta}) = -\frac{\partial^2 \kappa(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = -\text{Cov}(\mathbf{X})$$

and the matrix  $\text{Cov}(\mathbf{X})$  is positive definite (Barndorff-Nielsen, 1978).

Silvey (1959) gives seven assumptions containing the mathematical conditions required to ensure the existence of a ML estimator  $\hat{\boldsymbol{\theta}}$  of  $\boldsymbol{\theta}$ , the vector of unknown parameters appearing in the probability function, when there are certain constraints  $h(\boldsymbol{\theta}) = \mathbf{0}$  imposed on  $\boldsymbol{\theta}$ . Brown (1986) provides a comprehensive discussion of the analytic properties of the exponential family. The properties of the exponential family satisfy the assumptions of Silvey and this ensures the existence of the ML estimate of the mean of the exponential distribution when there are constraints of the form  $g(\boldsymbol{\mu}) = \mathbf{0}$ .

**Proposition 1**

Consider a random vector  $\mathbf{X}$ , with probability function belonging to the exponential family. Let  $\mathbf{g}(\mu)$  be a continuous vector-valued function of  $\mu$ ,

for which the first-order partial derivatives exist. Let  $\mathbf{G}_\mu = \frac{\partial \mathbf{g}(\mu)}{\partial \mu}$  be the derivative of  $\mathbf{g}(\mu)$  with respect to  $\mu$  and  $\mathbf{G}_z = \left. \frac{\partial \mathbf{g}(\mu)}{\partial \mu} \right|_{\mu=\mathbf{z}}$ .

The ML estimate of  $\mu$  subject to the constraints  $\mathbf{g}(\mu) = \mathbf{0}$ , is given by

$$(2.2) \quad \hat{\mu}_c = \mathbf{z} - (\mathbf{G}_\mu \mathbf{V})' (\mathbf{G}_z \mathbf{V} \mathbf{G}_\mu')^{-1} \mathbf{g}(\mathbf{z}) + o(\|\mathbf{z} - \mu\|).$$

This result implies that the ML estimate must be obtained iteratively.

**Proof:**

Let  $\gamma$  be a vector of Lagrange multipliers. To find the ML estimate of  $\mu$  subject to the constraints  $\mathbf{g}(\mu) = \mathbf{0}$ , we maximize

$$w(\mathbf{x}; \theta, \gamma) = \ln b(\mathbf{x}) + \mathbf{x}' \theta - \kappa(\theta) + \gamma' \mathbf{g}(\mu(\theta)).$$

Hence we find

$$\frac{\partial}{\partial \mu} w(\mathbf{x}; \theta, \gamma) = \frac{\partial}{\partial \theta} w(\mathbf{x}; \theta, \gamma) \left[ \frac{\partial \theta}{\partial \mu} \right].$$

Since we set  $\frac{\partial}{\partial \mu} w(\mathbf{x}; \theta, \gamma) = 0$  for a maximum, and since  $\left[ \frac{\partial \theta}{\partial \mu} \right]$  is invertible for a regular exponential family, we need further only consider  $\frac{\partial}{\partial \theta} w(\mathbf{x}; \theta, \gamma)$ .

Thus

$$\begin{aligned} \frac{\partial}{\partial \theta} w(\mathbf{x}; \theta, \gamma) &= \mathbf{x} - \frac{\partial}{\partial \theta} \kappa(\theta) + \frac{\partial}{\partial \theta} \{ \gamma' \mathbf{g}(\mu(\theta)) \} \\ &= \mathbf{x} - \mu + \left\{ \frac{\partial}{\partial \mu} \mathbf{g}(\mu(\theta)) \left[ \frac{\partial \mu}{\partial \theta} \right] \right\}' \gamma \\ &= \mathbf{x} - \mu + \left[ \frac{\partial \mu}{\partial \theta} \right]' \mathbf{G}_\mu' \gamma. \end{aligned}$$

Setting  $\frac{\partial}{\partial \theta} w(\mathbf{x}; \theta, \gamma) = 0$ , we get

$$(2.3) \quad \mu = \mathbf{x} + \left[ \frac{\partial \mu}{\partial \theta} \right]' \mathbf{G}_\mu' \gamma.$$

Using the linear Taylor expansion of  $g(\mu)$  about  $\mathbf{x}$ , we get

$$\begin{aligned} g(\mu) &= g\left[\mathbf{x} + \left[\frac{\partial \mu}{\partial \theta}\right]' G_{\mu}' \gamma\right] \\ &= g(\mathbf{x}) + G_z \left[\frac{\partial \mu}{\partial \theta}\right]' G_{\mu}' \gamma + o(\|\mathbf{x} - \mu\|). \end{aligned}$$

Setting  $g(\mu) = 0$  and solving for  $\gamma$ , gives

$$\gamma = - \left[ G_z \left[\frac{\partial \mu}{\partial \theta}\right]' G_{\mu}' \right]^{-1} g(\mathbf{x}) + o(\|\mathbf{x} - \mu\|).$$

Substituting  $\gamma$  in (2.3) we get

$$\hat{\mu}_c = \mathbf{x} - \left[ G_{\mu} \left[\frac{\partial \mu}{\partial \theta}\right] \right]' \left[ G_z \left[\frac{\partial \mu}{\partial \theta}\right]' G_{\mu}' \right]^{-1} g(\mathbf{x}) + o(\|\mathbf{x} - \mu\|).$$

Now

$$\frac{\partial \mu_i}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \left\{ \frac{\partial \kappa(\theta)}{\partial \theta_i} \right\} = \frac{\partial^2 \kappa(\theta)}{\partial \theta_j \partial \theta_i}.$$

Hence

$$\frac{\partial \mu}{\partial \theta} = \left[ \frac{\partial \mu_i}{\partial \theta_j} \right] = \left[ \frac{\partial^2 \kappa(\theta)}{\partial \theta_j \partial \theta_i} \right]$$

and

$$\left[ \frac{\partial \mu}{\partial \theta} \right]' = \left[ \frac{\partial^2 \kappa(\theta)}{\partial \theta_i \partial \theta_j} \right] = \mathbf{V},$$

from the fact that  $\kappa(\theta)$  is twice differentiable at an interior point  $\theta_0 \in \mathcal{N}$ , (Brown, 1986, theorem 2.2), and the Hessian matrix  $\mathcal{H}\kappa(\theta)$ , with  $ij$ -th element  $\frac{\partial^2 \kappa(\theta)}{\partial \theta_j \partial \theta_i}$ , is symmetric (Magnus and Neudecker, 1991, p. 105).

Therefore

$$\hat{\mu}_c = \mathbf{x} - (G_{\mu} \mathbf{V})' (G_z \mathbf{V} G_{\mu}')^{-1} g(\mathbf{x}) + o(\|\mathbf{x} - \mu\|)$$

which is the required result.

### Estimation

The variance covariance matrix  $V$  could be known, or it could be some function of  $\mu$ , say  $V_\mu$ . The iterative use of the estimation procedure thus depends on the form of  $G_\mu$  and  $V_\mu$ . Consider, for example, the estimation procedure for the following cases.

- (i) Let  $\mu = (\mu_1, \dots, \mu_p)'$  and  $D_\mu$  denote a diagonal matrix with the elements of  $\mu$  on the principal diagonal.

Suppose that  $g(\mu) = A \ln(\mu)$  and  $V = D_\mu$ , then

$$G_\mu = \frac{\partial}{\partial \mu} A \ln(\mu) = A D_\mu^{-1} \text{ and } G_z = A D_z^{-1}.$$

From (2.2) the ML estimate reduces to

$$\hat{\mu}_c = x - A' (A D_z^{-1} A')^{-1} A \ln(x) + o(\|x - \mu\|).$$

Let  $\mu^{(r)}$  denote the  $r$ -th approximation for the ML estimate  $\hat{\mu}_c$ . At step  $r$  in the iterative process ( $r = 0, 1, 2, \dots$ ),

$$(2.4) \quad \mu^{(r+1)} = \mu^{(r)} - A' (A D_{\mu^{(r)}}^{-1} A')^{-1} A \ln(\mu^{(r)}),$$

where for  $r = 0$ ,  $\mu^{(0)} = x$ , the vector of data. The process then converges to the ML estimate  $\hat{\mu}_c$ .

- (ii) If  $g$  is a linear function of  $\mu$ , say  $g(\mu) = A\mu$  and  $V = D_\mu$ , then  $G_\mu = A = G_z$  and (2.2) yields

$$\hat{\mu}_c = x - (A D_\mu)' (A D_\mu A')^{-1} A x.$$

The iterative process at step  $r$  ( $r = 0, 1, 2, \dots$ ), is

$$(2.5) \quad \mu^{(r+1)} = x - (A D_{\mu^{(r)}})' (A D_{\mu^{(r)}} A')^{-1} A x,$$

where, as in (i),  $\mu^{(0)} = x$ . The process then converges to the ML estimate  $\hat{\mu}_c$ .

- (iii) In some cases, as in this simple illustration, the procedure implies a double iteration, namely over  $\mathbf{x}$  and  $\mu$ . Suppose the elements of  $\mathbf{X} : 3 \times 1$  are independent Poisson random variables with parameter vector  $\mu$  and that the observed vector  $\mathbf{x}' = (80, 15, 5)$ . The model to be fitted is  $\mu_i = \alpha\beta^{i-1}$ . The model implies the restriction

$$g(\mu) = \mu_1\mu_3 - \mu_2^2 = 0.$$

In this case

$$\mathbf{V} = \mathbf{D}_\mu, \mathbf{G}_\mu = (\mu_3, -2\mu_2, \mu_1), \mathbf{G}_\mu \mathbf{D}_\mu = (\mu_1\mu_3, -2\mu_2^2, \mu_1\mu_3)$$

and

$$\mathbf{G}_z \mathbf{D}_\mu \mathbf{G}_\mu' = (x_1 + x_3)\mu_1\mu_3 + 4x_2\mu_2^2.$$

In this example the ML estimate of  $\mu$  must be found iteratively from

$$\hat{\mu}_c = \mathbf{x} - (\mathbf{G}_\mu \mathbf{D}_\mu)' (\mathbf{G}_z \mathbf{D}_\mu \mathbf{G}_\mu')^{-1} g(\mathbf{x}) + o(\|\mathbf{x} - \mu\|).$$

For every value of  $\mu$  the iteration is performed over  $\mathbf{x}$  to obtain a new estimate of  $\mu$ . This implies a double iteration over both  $\mathbf{x}$  and  $\mu$ .

- Both  $\mathbf{x}$  and  $\mu$  are initially estimated by the data. Now iterate over  $\mathbf{x}$  until convergence is attained. The values for  $\hat{\mu}_c$  during this first stage of iteration, are

$$\begin{bmatrix} 78,5263 \\ 16,6579 \\ 3,5263 \end{bmatrix} \text{ and } \begin{bmatrix} 78,5311 \\ 16,6525 \\ 3,5311 \end{bmatrix}.$$

- The last value  $\begin{bmatrix} 78,5311 \\ 16,6525 \\ 3,5311 \end{bmatrix}$  now becomes the second estimate of  $\mu$  in  $\mathbf{G}_\mu$  and  $\mathbf{D}_\mu$  for the second stage of the iteration. Once again iterate over  $\mathbf{x}$  with initial  $\mathbf{x} = \begin{bmatrix} 80 \\ 15 \\ 5 \end{bmatrix}$ .

Repeating this procedure, convergence is attained after four stages of the iterative process. The final ML estimate is

$$\hat{\mu}_c = \begin{bmatrix} 78,8218 \\ 17,3564 \\ 3,8218 \end{bmatrix}.$$

The same ML estimate may be obtained if the function  $g(\mu)$  is replaced by some other equivalent restriction, for example

$$g(\mu) = \sqrt{\mu_1 \mu_3} - \mu_2 = 0 \quad \text{or} \quad g(\mu) = \ln(\mu_1 \mu_3) - 2 \ln(\mu_2) = 0.$$

In the last instance,  $\hat{\mu}_c$  does not depend on  $\mu$  and iteration is only over  $\mathbf{x}$ . The same ML estimate is obtained in the case of multinomial sampling. The only difference is that  $\mathbf{V}$  is now replaced by the multinomial covariance matrix.

### Proposition 2

The asymptotic covariance matrix of  $\hat{\mu}_c$  is given by

$$(2.6) \quad \Sigma_c = \mathbf{V}_\mu - (\mathbf{G}_\mu \mathbf{V}_\mu)' (\mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu')^{-1} \mathbf{G}_\mu \mathbf{V}_\mu.$$

Proof:

$$\hat{\mu}_c = \mathbf{x} - (\mathbf{G}_\mu \mathbf{V}_\mu)' (\mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu')^{-1} g(\mathbf{x}) + o(\|\mathbf{x} - \mu\|).$$

Now since  $\mathbf{x} \xrightarrow{P} \mu$  and  $\mathbf{G}_\mu \xrightarrow{P} \mathbf{G}_\mu$ ,  $\hat{\mu}_c \xrightarrow{d} (\mathbf{I} \mathbf{U}) \left[ g(\mathbf{x}) \right]$ , where  $\mathbf{U} = -(\mathbf{G}_\mu \mathbf{V}_\mu)' (\mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu')^{-1}$ . Making use of the "delta method" (Bishop et al. (1975)), the asymptotic covariance matrix of  $\left[ g(\mathbf{x}) \right]$  is given by

$$\begin{aligned} & \left[ \frac{\partial}{\partial \mathbf{x}} \left[ g(\mathbf{x}) \right] \Big|_{\mathbf{x}=\mu} \right] \mathbf{V}_\mu \left[ \frac{\partial}{\partial \mathbf{x}} \left[ g(\mathbf{x}) \right] \Big|_{\mathbf{x}=\mu} \right]' \\ &= \begin{bmatrix} \mathbf{I} \\ \mathbf{G}_\mu \end{bmatrix} \mathbf{V}_\mu \begin{bmatrix} \mathbf{I} & \mathbf{G}_\mu' \end{bmatrix} = \begin{bmatrix} \mathbf{V}_\mu & (\mathbf{G}_\mu \mathbf{V}_\mu)' \\ \mathbf{G}_\mu \mathbf{V}_\mu & \mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu' \end{bmatrix}. \end{aligned}$$

Thus  $\hat{\mu}_c$  has asymptotic covariance matrix

$$\begin{aligned}\Sigma_c &= (\mathbf{I} \mathbf{U}) \begin{bmatrix} \mathbf{V}_\mu & (\mathbf{G}_\mu \mathbf{V}_\mu)' \\ \mathbf{G}_\mu \mathbf{V}_\mu \mathbf{V}_\mu & \mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu' \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{U}' \end{bmatrix} \\ &= \mathbf{V}_\mu - (\mathbf{G}_\mu \mathbf{V}_\mu)' (\mathbf{G}_\mu \mathbf{V}_\mu \mathbf{G}_\mu')^{-1} (\mathbf{G}_\mu \mathbf{V}_\mu).\end{aligned}$$

A useful application of the result in (2.2), which extends to ML estimation of the parameters in loglinear models, is the Poisson distribution.

**Example 1:**

Let  $X_i$ ,  $i = 1, \dots, p$  be independent Poisson variables with  $E(X_i) = \mu_i$ . The likelihood function is

$$l = \exp \left[ - \sum_{i=1}^p \mu_i \right] \prod_{i=1}^p \mu_i^{x_i} / \prod_{i=1}^p x_i!.$$

Thus

$$\ln l = - \sum_{i=1}^p \mu_i + \sum_{i=1}^p x_i \ln \mu_i - \ln \prod_{i=1}^p x_i!.$$

Hence  $\theta_i = \ln \mu_i$  and

$$\frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{\mu_i}, \quad i = 1, \dots, p.$$

Thus

$$\frac{\partial \theta}{\partial \mu} = \begin{pmatrix} \frac{1}{\mu_1} & & 0 \\ & \frac{1}{\mu_2} & \\ 0 & & \ddots & \\ & & & \frac{1}{\mu_p} \end{pmatrix}$$

and

$$\left[ \frac{\partial \theta}{\partial \mu} \right]^{-1} = D_{\mu}$$

where  $D_{\mu}$  is a diagonal matrix with the elements of  $\mu = (\mu_1, \dots, \mu_p)'$  on the principal diagonal. Thus from (2.2) the ML estimate of  $\mu = E(X)$  subject to some constraints  $g(\mu) = 0$  is

$$(2.7) \quad \hat{\mu}_c = \bar{x} - (G_{\mu} D_{\mu})' (G_{\mu} D_{\mu} G_{\mu}')^{-1} g(\bar{x}) + o(\|\bar{x} - \mu\|).$$

### Example 2:

If  $X : p \times 1$  is  $N(\mu, \Sigma)$  then Barndorff-Nielsen (1978) gives the canonical parameters which are functions of  $\mu$ , as  $\theta = \Sigma^{-1} \mu$ .

Thus

$$\frac{\partial \theta}{\partial \mu} = \Sigma^{-1} \quad \text{and} \quad \frac{\partial \mu}{\partial \theta} = \Sigma.$$

Let  $G$  denote a matrix of known constants. Then the ML estimate of  $\mu$  subject to the constraints  $G\mu = 0$ , is given by

$$\hat{\mu}_c = \bar{x} - (G\Sigma)' (G\Sigma G')^{-1} G\bar{x},$$

where  $\bar{x}$  is a ML estimate of  $\mu$ . This is the result given in theorem 1 of Crowther and Shaw (1989), which is derived using the conditional distribution of the multivariate normal distribution.

### 3. APPLICATIONS

The estimation procedure may be applied to a wide variety of problems. It proves to be a very useful and natural way of modelling data when constraints are linear as well as non-linear.

The approach accommodates any choice of model assuming any predetermined distribution of the error terms involved, provided that the covariance matrix of the error terms can be computed as in the case of the generalized linear model. Other constraints such as forcing regression functions through fixed points may be incorporated.

The method can easily be implemented in analyzing categorical data

resulting from sampling schemes such as the Poisson, multinomial, negative binomial, hypergeometric or any complex sampling scheme and in cases where weighting of the data is essential.

Some of the applications are considered below.

### 3.1 Estimating a covariance matrix under constraints

The procedure may be applied to estimate the covariance matrix of, for example, a multivariate normal distribution under certain constraints. An example would be to estimate the variances, given that the correlation coefficients are known values. Consider the bivariate case where the covariance matrix  $\Sigma = (\sigma_{ij})$  is estimated by

$\frac{1}{n}A$  and  $A = (a_{ij})$  has a Wishart distribution. The estimation problem is that of estimating  $\sigma = (\sigma_{11}, \sigma_{22}, \sigma_{12})'$  given that  $\rho = c$ .

The covariance matrix of  $\mathbf{x} = \frac{1}{n}(a_{11}, a_{22}, a_{12})'$  is

$$V = \frac{1}{n} \begin{pmatrix} 2\sigma_{11} & 2\sigma_{22} & 2\sigma_{11}\sigma_{12} \\ 2\sigma_{12} & 2\sigma_{22} & 2\sigma_{22}\sigma_{12} \\ 2\sigma_{11}\sigma_{12} & 2\sigma_{22}\sigma_{12} & \sigma_{11}\sigma_{22} + \sigma_{12}^2 \end{pmatrix}.$$

In this case

$$g(\sigma) = \sigma_{12} - c\sqrt{\sigma_{11}\sigma_{22}} \text{ and } G_{\sigma} = \left[ -\frac{1}{2}c\sqrt{\sigma_{22}/\sigma_{11}}, -\frac{1}{2}c\sqrt{\sigma_{11}/\sigma_{22}}, 1 \right]$$

The ML estimate of  $\sigma$  subject to the constraint  $\rho = c$  may be obtained from:

$$\hat{\sigma}_c = \mathbf{x} - (G_{\sigma}'V)(G_{\sigma}'VG_{\sigma}')^{-1}g(\mathbf{x}) + o(\|\mathbf{x} - \sigma\|).$$

If the special case of  $n = 100$  and  $\frac{1}{n}(a_{11}, a_{22}, a_{12}) = (3, 3, 2)$  is considered, the ML estimates of the corresponding elements of  $\Sigma$  under the restriction  $\rho = 0.5$  are  $(2.667, 2.667, 1.333)$ , which are found after using the prescribed double iteration.

### 3.2 The loglinear model

Another useful application is the loglinear model where, for example, a Poisson sampling procedure is used, namely that the frequency of each cell in the contingency table is an independent observation of a Poisson random variable  $X_i$  with mean  $F_i$  for cell  $i$ . If the frequencies for the cells are conveniently arranged into a vector, say  $\mathbf{x} = (x_1, \dots, x_p)'$ , the saturated loglinear model can be written as

$$(3.1) \quad \ell n(\mathbf{F}) = \mathbf{A}\lambda$$

where  $\mathbf{F} = (F_1, F_2, \dots, F_p)'$  is the mean vector for the Poisson random variables,  $\mathbf{A} : p \times p$  is the design matrix and  $\lambda : p \times 1$  is the vector of parameters for the mean, main effects and interaction effects. Fitting a lower order model implies fitting a model where certain elements of  $\lambda$  are zero. The saturated model of (3.1) can be written as

$$(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\ell n(\mathbf{F}) = \lambda.$$

The hypothesis that certain linear functions of  $\lambda$  are zero, can be written in the form

$$\mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\ell n(\mathbf{F}) = \mathbf{H}\lambda = \mathbf{0},$$

where  $\mathbf{H}$  is a matrix specifying the linear functions of  $\lambda$  set to zero.

Let  $\mathbf{A}'_H = \mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$ . We now have a function

$$\mathbf{g}(\mathbf{F}) = \mathbf{A}'_H \ell n(\mathbf{F}) = \mathbf{0}.$$

The ML estimate of  $\mathbf{F}$  subject to the constraints  $\mathbf{g}(\mathbf{F}) = \mathbf{A}'_H \ell n(\mathbf{F}) = \mathbf{0}$ , is given by (2.7), where

$$\mathbf{G}_F = \frac{\partial}{\partial \mathbf{F}} \mathbf{g}(\mathbf{F}) = \mathbf{A}'_H \mathbf{D}_F^{-1}.$$

Thus

$$\begin{aligned}
 (3.2) \quad \hat{\mathbf{F}}_c &= \mathbf{x} - (\mathbf{A}'_H \mathbf{D}_F^{-1} \mathbf{D}_F)' (\mathbf{A}'_H \mathbf{D}_F^{-1} \mathbf{D}_F \mathbf{D}_F^{-1} \mathbf{A}_H)^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|) \\
 &= \mathbf{x} - \mathbf{A}_H (\mathbf{A}'_H \mathbf{D}_F^{-1} \mathbf{A}_H)^{-1} \mathbf{A}'_H \ell_n(\mathbf{X}) + o(\|\mathbf{x} - \mathbf{F}\|).
 \end{aligned}$$

$\hat{\mathbf{F}}_c$  must be determined iteratively. Let  $\mu^{(r)}$  denote the  $r$ -th approximation for the ML estimate  $\hat{\mathbf{F}}_c$ , then at step  $r$  in the iterative process ( $r = 0, 1, 2, \dots$ )

$$\mu^{(r+1)} = \mu^{(r)} - \mathbf{A}_H (\mathbf{A}'_H \mathbf{D}_{\mu^{(r)}}^{-1} \mathbf{A}_H)^{-1} \mathbf{A}'_H \ell_n(\mu^{(r)}),$$

where for  $r = 0$ ,  $\mu^{(0)} = \mathbf{x}$ , the original frequency vector. The procedure then converges to the ML estimate  $\hat{\mathbf{F}}_c$ .

The ML estimates of the parameters in the loglinear model are given by

$$(3.3) \quad \hat{\lambda}_c = (\mathbf{A}' \mathbf{A})^{-1} \mathbf{A}' \ell_n(\hat{\mathbf{F}}_c).$$

The asymptotic covariance matrix of  $\hat{\mathbf{F}}_c$  can be obtained from (2.6), where  $\mathbf{V}_\mu$  is replaced by  $\text{Cov}(\mathbf{X}) = \mathbf{D}_F$  and is given by

$$(3.4) \quad \Sigma_c = \mathbf{D}_F - \mathbf{A}_H (\mathbf{A}'_H \mathbf{D}_F^{-1} \mathbf{A}_H)^{-1} \mathbf{A}'_H.$$

The estimated asymptotic covariance matrix of  $\hat{\mathbf{F}}_c$  is given by

$$(3.5) \quad \hat{\Sigma}_c = \mathbf{D}_{\hat{\mathbf{F}}_c} - \mathbf{A}_H (\mathbf{A}'_H \mathbf{D}_{\hat{\mathbf{F}}_c}^{-1} \mathbf{A}_H)^{-1} \mathbf{A}'_H.$$

The covariance matrix for  $\hat{\lambda}_c$  is

$$\text{Cov}(\hat{\lambda}_c) = (\mathbf{A}' \mathbf{A})^{-1} \mathbf{A}' \text{Cov}[\ell_n(\hat{\mathbf{F}}_c)] \mathbf{A} (\mathbf{A}' \mathbf{A})^{-1},$$

where once again applying the "delta method", the estimated asymptotic covariance matrix

$$\begin{aligned} \text{est.}[Cov(\ln(\hat{F}_c))] &= \left[ \frac{\partial}{\partial \hat{F}_c} \ln(\hat{F}_c) \right] \hat{\Sigma}_c \left[ \frac{\partial}{\partial \hat{F}_c} \ln(\hat{F}_c) \right]' \\ &= D_{\hat{F}_c}^{-1} \hat{\Sigma}_c D_{\hat{F}_c}^{-1}. \end{aligned}$$

Hence the estimated covariance matrix for  $\lambda_c$  is

$$(3.6) \quad \text{est.}[Cov(\hat{\lambda}_c)] = (A' A)^{-1} A' [D_{\hat{F}_c}^{-1} \hat{\Sigma}_c D_{\hat{F}_c}^{-1}] A (A' A)^{-1}.$$

To illustrate the above-mentioned estimation procedure, consider the following applications in loglinear modelling using a Poisson sampling scheme.

### (i) Independence

Consider a  $3 \times 3$  contingency table with row variable  $B$  and column variable  $C$ . The independence model is written as  $\ln(F_{ij}) = \mu + \lambda_i^B + \lambda_j^C$ , which means testing

$$H_0 : \lambda_{ij}^{BC} = 0, \quad i, j = 1, 2;$$

in the saturated model.

Let  $\lambda' = [\mu, \lambda_1^B, \lambda_2^B, \lambda_1^C, \lambda_2^C, \lambda_{11}^{BC}, \lambda_{12}^{BC}, \lambda_{21}^{BC}, \lambda_{22}^{BC}]$ .

The design matrix is given by

$$A = \begin{bmatrix} \mu & \lambda_1^B & \lambda_2^B & \lambda_1^C & \lambda_2^C & \lambda_{11}^{BC} & \lambda_{12}^{BC} & \lambda_{21}^{BC} & \lambda_{22}^{BC} \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & -1 & 0 & 0 & -1 & -1 \\ 1 & -1 & -1 & 1 & 0 & -1 & 0 & -1 & 0 \\ 1 & -1 & -1 & 0 & 1 & 0 & -1 & 0 & -1 \\ 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Now let

$$\mathbf{H} = [0_{4 \times 5}, \mathbf{I}_4].$$

Then  $\mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}' \ell_n(\mathbf{F}) = \mathbf{H}\lambda = 0$ , gives

$$\begin{bmatrix} \lambda_{11}^{BC} \\ \lambda_{12}^{BC} \\ \lambda_{21}^{BC} \\ \lambda_{22}^{BC} \end{bmatrix} = 0,$$

which is the statement under  $H_o$ .  $\hat{\mathbf{F}}_c$  and  $\hat{\lambda}_c$  can now be found by using (3.2) and (3.3) with  $\mathbf{H}$  above.

## (ii) Symmetry

For symmetry in the  $3 \times 3$  contingency table,  $F_{ij} = F_{ji}$ , and this implies testing

$$H_o : \lambda_{ij}^{BC} = \lambda_{ji}^{BC} \text{ and } \lambda_i^B = \lambda_i^C, \quad i, j = 1, 2.$$

Consider

$$\mathbf{H} = \begin{bmatrix} \mu & \lambda_1^B & \lambda_2^B & \lambda_1^C & \lambda_2^C & \lambda_{11}^{BC} & \lambda_{12}^{BC} & \lambda_{21}^{BC} & \lambda_{22}^{BC} \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{bmatrix}.$$

Thus  $F_o$  can once again be written as

$$\mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}' \ell_n(\mathbf{F}) = \mathbf{H}\lambda = 0.$$

For quasi-symmetry  $\mathbf{H} = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ -1 \ 0]$ .

The ML estimate for the expected frequency  $\hat{\mathbf{F}}_c$  is found by iteratively using (3.2), and  $\hat{\lambda}_c$  is given by (3.3).

### 3.3 Logits, cumulative logits and variations

The logit, cumulative logit model and other variations such as the adjacent-categories logit model of Agresti (1984), can all be written in terms of the saturated model as  $\mathbf{K} \ell_n(\mathbf{CF}) = \mathbf{A}\lambda$ , where  $\mathbf{K}$  and  $\mathbf{C}$  are appropriately defined for the different models. A lower order model can once again be found by setting certain elements of  $\lambda$  equal to zero. This can be done by writing

$$\mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{K} \ell_n(\mathbf{CF}) = \mathbf{H}\lambda = \mathbf{0},$$

or  $\mathbf{A}'_H \mathbf{K} \ell_n(\mathbf{CF}) = \mathbf{H} \lambda = \mathbf{0}$ , where  $\mathbf{A}'_H = \mathbf{H}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$ . Thus  $g(\mathbf{F}) = \mathbf{A}'_H \mathbf{K} \ell_n(\mathbf{CF})$  and  $\mathbf{G} = \frac{\partial g(\mathbf{F})}{\partial \mathbf{F}} = \mathbf{A}'_H \mathbf{K} \mathbf{D}_{\mathbf{CF}}^{-1} \mathbf{C}$ , where  $\mathbf{D}_{\mathbf{CF}}$  is a diagonal matrix with the elements of  $\mathbf{CF}$  on the principal diagonal. The ML estimates for the expected frequencies,  $\hat{\mathbf{F}}_c$  for the model, can be found by using (2.2).

### 3.4 Ordinal models

The ordinal models of Agresti (1984, 1990) can also be fitted using this estimation procedure. The ordinal model can be written as  $\ell_n(\mathbf{F}) = \mathbf{X}\beta$ , where  $\mathbf{X}$  has main effects and some covariates as column vectors, instead of the standard interaction terms. Let  $\mathbf{P} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ , then  $\mathbf{P} \ell_n(\mathbf{F}) = \mathbf{P}\mathbf{X}\beta = \mathbf{0}$ . Thus  $g(\mathbf{F}) = \mathbf{P} \ell_n(\mathbf{F})$  and  $\mathbf{G} = \mathbf{P}\mathbf{D}_F^{-1}$ . Thus (2.2) can be used iteratively for estimation, with  $\mathbf{V} = \mathbf{D}_z$  in the first iteration. The logit models with one or more ordinal predictor variables, can be written as  $\mathbf{K} \ell_n(\mathbf{CF}) = \mathbf{X}\beta$  with  $\mathbf{X}$  the appropriate design matrix.  $\mathbf{P}$  is chosen as above and  $\hat{\mathbf{F}}_c$  can be determined iteratively.

## 4. COMPUTATIONS

The estimated parameters may be easily computed using a matrix algebra package, for example the SAS/IML procedure of the SAS System. Consider the following examples with output from the IML programs.

**Example 1: The Poisson regression problem**

For the Poisson regression model, we have  $\mu = E(Y) = X\beta$  where  $Cov(Y) = D_\mu$ .

Let  $P = I - X(X'X)^{-1}X'$ . Then  $P\mu = PX\beta = 0$ . Thus  $g(\mu) = P\mu$  and  $G = \frac{\partial g(\mu)}{\partial \mu} = P$ . The ML estimate of the mean in the model is found by applying (2.5), for which the iterative process at step  $r$  ( $r = 0, 1, 2, \dots$ ) is

$$\mu^{(r+1)} = y - (PD_{\mu^{(r)}})'(PD_{\mu^{(r)}}P')^{-1}Py,$$

where  $\mu^{(0)} = y$ , and once again the process converges to the ML estimate.

Consider the following example from Dobson (1983). The data are counts  $y_i$  observed at various values of a covariate  $x_i$ .

|       |    |    |   |   |   |   |    |    |    |
|-------|----|----|---|---|---|---|----|----|----|
| $y_i$ | 2  | 3  | 6 | 7 | 8 | 9 | 10 | 12 | 15 |
| $x_i$ | -1 | -1 | 0 | 0 | 0 | 0 | 1  | 1  | 1  |

Observing that the variability increases with  $x$ , assume that the responses  $Y_i$  are Poisson random variables with

$$E(Y_i) = \mu_i = \beta_0 + \beta_1 x_i, \quad i = 1, \dots, 9.$$

Dobson uses the iterative reweighted least squares procedure to find the ML estimates of the parameters. Using the procedure above, the ML estimates are  $b_0 = 7.4516$  and  $b_1 = 4.9353$ , which agree with those obtained by Dobson.

The next example illustrates the fitting of a cumulative logit model. Let  $X$  and  $Y$  be explanatory variables while  $Z$  denotes an ordinal response variable. Suppose that there are  $r \times c$  subpopulations formed by the category combinations of  $X$  and  $Y$ . At levels  $i$  of  $X$  and  $j$  of  $Y$  and for cut point  $\ell$ , the cumulative logit is defined as follows,

$$L_{\ell(ij)} = \ell n \left[ \frac{F_{ij1} + \dots + F_{ij\ell}}{F_{ij\ell+1} + \dots + F_{ijk}} \right], \quad \ell = 1, \dots, k-1.$$

**Example 2: A cumulative logit model**

In a survey on socio-political change in South Africa of October 1987 people were asked their opinion as to how effectively the unrest situation had been handled. The results are summarized in the following contingency table.

**TABLE I**  
**OBSERVED FREQUENCIES AND EXPECTED FREQUENCIES FOR**  
**CUMULATIVE LOGIT MODEL FOR OPINION**  
**ON THE UNREST SITUATION**

| Language             | Age   | Opinion     |                                      |                     |                   |
|----------------------|-------|-------------|--------------------------------------|---------------------|-------------------|
|                      |       | Ineffective | Neither effective<br>nor ineffective | Fairly<br>effective | Very<br>effective |
| Afrikaans            | 18-24 | 5 (5,33)    | 6 (8,25)                             | 53 (53,25)          | 16 (13,17)        |
|                      | 25-34 | 7 (7,95)    | 15 (12,90)                           | 115 (106,43)        | 25 (34,72)        |
|                      | 35-64 | 20 (11,72)  | 20 (19,73)                           | 197 (205,21)        | 96 (89,35)        |
| English<br>and other | 18-24 | 7 (5,49)    | 3 (4,58)                             | 33 (32,48)          | 7 (7,45)          |
|                      | 25-34 | 8 (8,60)    | 9 (7,59)                             | 65 (68,33)          | 23 (20,48)        |
|                      | 35-64 | 13 (13,98)  | 13 (12,91)                           | 149 (146,11)        | 56 (58,00)        |

(Figures in parentheses are the expected frequencies for the cumulative logit model  $L_{(ij)} = \mu_{\ell} + \tau_{i\ell}^T + \beta_{\ell}^A(v_j - \bar{v})$ ,  $\ell = 1, 2, 3$ ).

Consider "opinion" as the response variable, with "language" a nominal explanatory variable and "age" an ordinal explanatory variable. The non-homogeneous model for the data is

$$L_{(ij)} = \mu_{\ell} + \tau_{i\ell}^T + \beta_{\ell}^A(v_j - \bar{v}), \quad \ell = 1, 2, 3.$$

When this model is fitted to the data, it is found that the  $\hat{\beta}_k^A$  are similar in magnitude ( $\hat{\beta}_1^A = -0,3656$ ,  $\hat{\beta}_2^A = -0,2747$  and  $\hat{\beta}_3^A = -0,3555$ ), hence the following model is fitted to the data

$$L_{(ij)} = \mu_{\ell} + \tau_{i\ell}^L + \beta^A(v_j - \bar{v}), \quad \ell = 1, 2, 3.$$

The parameter estimates and their standard normal values are given in table II.

TABLE II

| parameter   | estimate | z-value |
|-------------|----------|---------|
| $\mu_1$     | -2,6908  | -18,59  |
| $\tau_{11}$ | -0,2737  | -1,93   |
| $\beta^A$   | -0,3249  | -3,29   |
| $\mu_2$     | -1,8072  | -17,74  |
| $\tau_{12}$ | -0,1052  | -1,08   |
| $\mu_3$     | 1,3584   | 14,59   |
| $\tau_{13}$ | -0,0594  | -0,75   |

The cumulative logit model fits the data well. The likelihood-ratio statistic,  $LR = 8,35$  with  $df = 11$ .

### Example 3: Marginal homogeneity

Consider the problem of comparing two marginal distributions of a square table. Marginal homogeneity states that  $\pi_{i+} = \pi_{+i}$ ,  $i = 1, \dots, r$ .

As an example consider the well-known "unaided vision data set" of Bhapkar (1966) also discussed by Bishop et al. (1975). The observed frequencies and ML estimates for the expected frequencies (in parentheses), for the marginal homogeneity model are shown in table III. The hypothesis of marginal homogeneity is  $H_0: \pi_{i+} = \pi_{+i}$ ,  $i = 1, 2, 3, 4$ .

Let  $\pi' = (\pi_{11}, \pi_{12}, \pi_{13}, \pi_{14}, \pi_{21}, \dots, \pi_{24}, \dots, \pi_{41}, \dots, \pi_{44})$ .

Notice that  $\pi_{1+} = \pi_{+1}$  gives

$$\pi_{11} + \pi_{12} + \pi_{13} + \pi_{14} = \pi_{11} + \pi_{21} + \pi_{31} + \pi_{41}.$$

TABLE III

UNAIDED DISTANCE VISION FOR 7477 WOMEN AGED 30-39

| Right eye     | Left eye      |              |              |              | Total |
|---------------|---------------|--------------|--------------|--------------|-------|
|               | Highest grade | Second grade | Third grade  | Lowest grade |       |
| Highest grade | 1520 (1520)   | 266 (252,48) | 124 (111,84) | 66 (56,97)   | 1976  |
| Second grade  | 234 (247,24)  | 1512 (1512)  | 432 (409,42) | 78 (70,59)   | 2256  |
| Third grade   | 117 (131,27)  | 362 (383,13) | 1772 (1772)  | 205 (195,26) | 2456  |
| Lowest grade  | 36 (42,79)    | 82 (91,63)   | 179 (188,40) | 492 (492)    | 789   |
| Total         | 1970          | 2222         | 2507         | 841          | 7477  |

$$\text{i.e. } \pi_{12} + \pi_{13} + \pi_{14} - \pi_{21} - \pi_{31} - \pi_{41} = 0$$

which can be written as

$$(0 \ 1 \ 1 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)\pi = 0.$$

Furthermore  $H_o$  can be written as  $A\pi = 0$ , where

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Since  $A$  is singular, any 3 rows of  $A$  may be used to represent  $H_o$  and thus 3 degrees of freedom are associated with  $H_o$ .

The constraints may be written as  $g(\pi) = A\pi = 0$  or  $g(F) = AF = 0$ .

Hence

$$G = \frac{\partial}{\partial F} g(F) = A.$$

Thus the estimated expected frequency is found iteratively using (2.5), where at step  $r$  in the iterative procedure ( $r = 0, 1, 2, \dots$ ),

$$\mu^{(r+1)} = \mathbf{x} - (\text{AD}_{\mu^{(r)}})' (\text{AD}_{\mu^{(r)}} \mathbf{A}')^{-1} \mathbf{A} \mathbf{x},$$

where,  $\mu^{(0)} = \mathbf{x}$  is the original frequency vector. Applying the above estimation procedure gives ML estimates of the expected frequencies, shown in parentheses in table III, with  $\chi^2 = 11.97$  and  $LR = 11.99$  with  $df = 3$ .

## 5. CONCLUSION

The estimation procedure gives a maximum likelihood method which is very flexible for modelling, and can be applied to a wide variety of problems, where the approach is to view the model in terms of the implied constraints  $g(\mu) = 0$ , as illustrated by the examples.

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